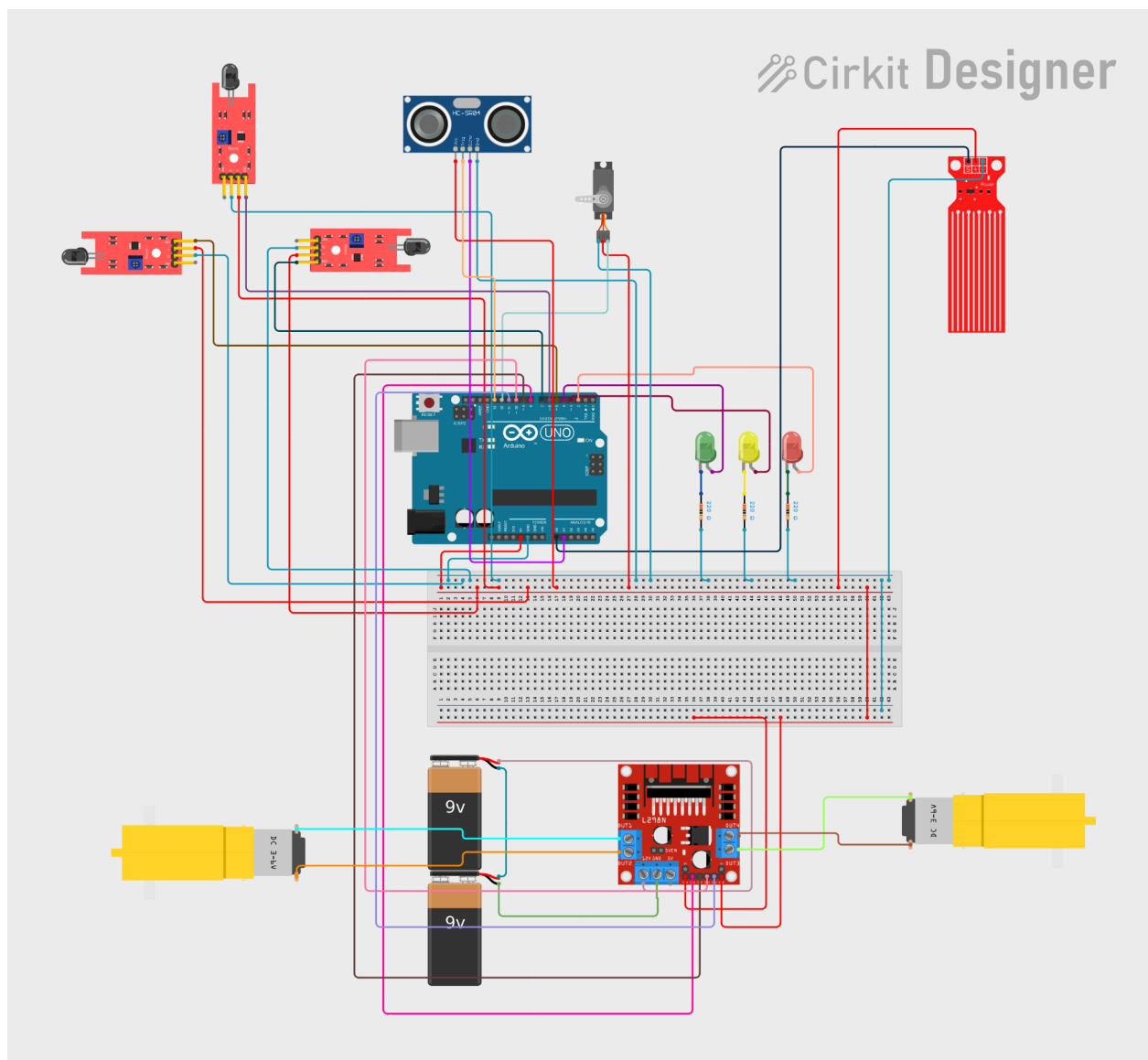


Circuit Documentation

Summary

This document provides a detailed overview of a multi-functional circuit designed to detect water levels, sense flames, and control a robotic vehicle with a water valve. The circuit utilizes an Arduino UNO microcontroller to interface with various sensors and actuators, including LEDs, flame sensors, a water level sensor, a servo motor, DC gear motors, and an ultrasonic sensor. The system is powered by a combination of a 9V battery and the Arduino's 5V output.



Component List

1. **Arduino UNO**
 - A microcontroller board based on the ATmega328P, used to control the entire circuit.
2. **LED Two Pin (Green)**
 - A green LED used to indicate high water levels.
3. **LED Two Pin (Red)**
 - A red LED used to indicate low water levels.
4. **LED Two Pin (Yellow)**
 - A yellow LED used to indicate medium water levels.
5. **Resistor (220 Ohms)**
 - Used to limit current through the LEDs.
6. **Water Level Sensor**
 - Detects the water level and provides an analog signal to the Arduino.
7. **Servo**
 - Controls a water valve, opening or closing it based on fire detection and water level.
8. **Flame Sensor**
 - Detects the presence of fire and provides a digital signal to the Arduino.
9. **L298N Motor Driver**
 - Controls the DC gear motors for vehicle movement.
10. **9V Battery**
 - Provides power to the motor driver and motors.
11. **DC Gear Motor (3-6V)**
 - Drives the robotic vehicle.
12. **HC-SR04 Ultrasonic Sensor**
 - Measures distance to detect proximity to fire.

Wiring Details

Arduino UNO

- **5V:** Connected to VCC of Water Level Sensor, Servo, Flame Sensors, Ultrasonic Sensor, and ENA/ENB of L298N Motor Driver.
- **GND:** Connected to GND of Resistors, Water Level Sensor, Servo, Flame Sensors, and Ultrasonic Sensor.
- **D2:** Connected to Anode of Red LED.
- **D3:** Connected to Anode of Yellow LED.
- **D4:** Connected to Anode of Green LED.
- **A0:** Connected to SIG of Water Level Sensor.
- **D12:** Connected to PWM of Servo.
- **D5:** Connected to D0 of Left Flame Sensor.
- **D6:** Connected to D0 of Center Flame Sensor.
- **D7:** Connected to D0 of Right Flame Sensor.
- **D8:** Connected to IN1 of L298N Motor Driver.

- **D9:** Connected to IN2 of L298N Motor Driver.
- **D10:** Connected to IN3 of L298N Motor Driver.
- **D11:** Connected to IN4 of L298N Motor Driver.
- **A1:** Connected to ECHO of Ultrasonic Sensor.
- **D13:** Connected to TRIG of Ultrasonic Sensor.

LED Two Pin (Green)

- **Anode:** Connected to D4 of Arduino UNO.
- **Cathode:** Connected to pin1 of a 220 Ohm Resistor.

LED Two Pin (Red)

- **Anode:** Connected to D2 of Arduino UNO.
- **Cathode:** Connected to pin1 of a 220 Ohm Resistor.

LED Two Pin (Yellow)

- **Anode:** Connected to D3 of Arduino UNO.
- **Cathode:** Connected to pin1 of a 220 Ohm Resistor.

Resistors (220 Ohms)

- **pin1:** Connected to Cathode of respective LEDs.
- **pin2:** Connected to GND of Arduino UNO.

Water Level Sensor

- **SIG:** Connected to A0 of Arduino UNO.
- **VCC:** Connected to 5V of Arduino UNO.
- **GND:** Connected to GND of Arduino UNO.

Servo

- **PWM:** Connected to D12 of Arduino UNO.
- **VCC:** Connected to 5V of Arduino UNO.
- **GND:** Connected to GND of Arduino UNO.

Flame Sensors

- **D0:** Connected to respective digital pins (D5, D6, D7) of Arduino UNO.
- **VCC:** Connected to 5V of Arduino UNO.
- **GND:** Connected to GND of Arduino UNO.

L298N Motor Driver

- **12V:** Connected to Positive of 9V Battery.
- **GND:** Connected to Negative of 9V Battery.

- **ENA/ENB:** Connected to 5V of Arduino UNO.
- **IN1, IN2, IN3, IN4:** Connected to D8, D9, D10, D11 of Arduino UNO.
- **OUT1, OUT2:** Connected to V- and V+ of one DC Gear Motor.
- **OUT3, OUT4:** Connected to V+ and V- of another DC Gear Motor.

9V Battery

- **Positive:** Connected to 12V of L298N Motor Driver.
- **Negative:** Connected to Positive of another 9V Battery.

DC Gear Motors

- **V+ and V-:** Connected to OUT1, OUT2, OUT3, OUT4 of L298N Motor Driver.

HC-SR04 Ultrasonic Sensor

- **TRIG:** Connected to D13 of Arduino UNO.
- **ECHO:** Connected to A1 of Arduino UNO.
- **VCC:** Connected to 5V of Arduino UNO.
- **GND:** Connected to GND of Arduino UNO.

Code Documentation

The Arduino code is responsible for controlling the various components based on sensor inputs. Below is a brief overview of the code functionality:

- **Water Level Detection:** The water level sensor provides an analog input to the Arduino, which is used to control the LEDs. The red LED indicates low water levels, the yellow LED indicates medium levels, and the green LED indicates high levels.
- **Flame Detection:** The flame sensors provide digital inputs to the Arduino. If any sensor detects a flame, the system checks the distance using the ultrasonic sensor. If the flame is within a safe distance, the water valve is opened, and the motors are stopped.
- **Motor Control:** The L298N motor driver is used to control the DC gear motors. The motors are controlled based on the flame sensor inputs to move the vehicle towards the flame or stop it.
- **Servo Control:** The servo motor is used to control a water valve. The valve is opened or closed based on the flame detection and water level.
- **Ultrasonic Distance Measurement:** The ultrasonic sensor measures the distance to the flame. If the flame is within a specified safe distance, the water valve is opened.

The code is structured with functions to handle each of these tasks, ensuring modularity and ease of understanding. The main loop continuously checks sensor inputs and updates the system state accordingly.